

Jonathan Bourget

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📅 Available January 2027 · 12-month co-op work term

Education

B.Eng., Electrical Engineering (Co-op) — Carleton University

Sept 2024 – Apr 2029 (expected)

- Third-Year Standing | GPA available on request | Carleton Academic Scholarship (2024–26)

Hardware & Systems Projects

Flight Electronics & Power Distribution

Electronics Lead · 2026 – Present

- Architecting the electrical system for a twin-motor RC aircraft ahead of the airframe build so the aero team designs around fixed power, mass, and wiring constraints; created and maintained the project **BOM** and **CPM/Gantt** schedule.
- Designed a 4-layer power distribution board in **Altium Designer**, splitting a 4S battery into raw motor power and regulated 12V, 5V, and 3.3V rails to isolate motor, logic, and sensor loads.
- Peer review showed the linear 12V rail would drop out as the battery voltage sags under motor load; V2 moves to a **buck converter** and adds input spike suppression, per-regulator decoupling, and copper-pour current paths.
- Brought up a **2.4 GHz ExpressLRS** link and decoded all 16 control channels on **STM32** over CRSF/UART (library-based).
- Bench-testing a BNO085 9-axis IMU on the STM32 flight controller ahead of flight-control firmware development.

Self-Hosted AI Secretary & LED Wall Display

Solo Build · 2026 – Present

- Architectured an assistant on an **Orange Pi Zero 3** (ARM Linux) whose Node.js/Express and SQLite backend merges Gmail, Google Calendar, course deadlines, and live prices for 40+ stock holdings into one prioritized daily brief.
- Designed an **ESP32-S3 HUB75 LED wall** (192×32 px) that polls the Pi at two rates so live controls respond in near real time.
- Deployed a React dashboard with an interactive and conversational interface over HTTPS via Tailscale; implemented with AI coding tools (Claude Code, Codex) to self-authored architecture and specifications.

RFID Feasibility Infrastructure Proposal

Team Study · 2026

- Evaluated **RFID**-based smart bike infrastructure for Carleton, comparing tag/reader technologies and cost against existing systems.
- Weighed user needs against feasibility constraints and delivered the findings as a formal proposal and executive presentation.

MOSFET-Switched Off-Road Lighting

Solo Build · 2025

- Retrofitted high-power roof-rack LEDs switched by **MOSFETs** from an in-dash Arduino on a fused circuit, with custom flash patterns.
- Sized parts from datasheets for –35 to 40 °C, peak 10A continuous and bench-tested load and heat before install; added weatherproof connectors.

Programmable Bicycle Lighting PCB

Solo Build · 2022

- Took a programmable RGBW Adafruit board from breadboard to a **custom Arduino-controlled PCB**.
- Designed three 3D-printed weatherproof enclosures (front, rear, compact) that offer modular install with varying PCB sizes.

Technical Proficiencies

PCB & Circuit Design	Altium Designer · KiCad · Schematic Capture · 4-Layer PCB Layout · Power Delivery · Component Selection
Embedded Systems	STM32 · ESP32 · Arduino · Orange Pi (Linux/ARM)
Programming	Python · C/C++ · Verilog · Java · Git · LaTeX
Design & Simulation	MATLAB/Simulink · Vivado (FPGA) · Autodesk Fusion · ANSYS Electronics
Interfaces & Protocols	UART/SPI/I2C · ExpressLRS · CRSF · RFID
Lab Equipment	Oscilloscope · Signal Generator · Multimeter · Bench Power Supplies
Fabrication	Soldering · 3D Printing · CAD · Welding
Languages	English · French (Bilingual)

Work Experience

Lifeguard, Swim Instructor & First Aid Instructor — City of Ottawa

2021 – Present

- **Teach and evaluate** swim lessons and first aid courses while working with candidates to overcome weaknesses.
- Make time-critical decisions in a high-liability environment, **coordinating team response** and maintaining rigorous incident documentation.
- Certified: Standard First Aid, CPR-C with AED, Airway Management & Oxygen Administration, SFA Instructor/Examiner